Database Systems

SQL Basics

SQL

- **SQL: Structured Query Language**
  - Data Definition Language (DDL)
    The **Data-Definition** sublanguage for declaring database schemas
  - Data Manipulation Language (DML)
    The **Data-Manipulation** sublanguage for querying databases and for modifying the database
- SQL
  - **Declarative** language
  - Supported by all major commercial database systems
  - Standardized: many new features over time
- SQL makes a distinction between
  - **Stored relations** (**Tables**): Exists in DB and can be modified and queried
  - **Views** (Defined by Computation): Not stored but constructed when needed
  - **Temporary tables**: Constructed by SQL processor when it performs its job of executing queries and data modifications

SQL: Data Types

- Primitive Data Types
  - Character string of fixed or varying Length (`CHAR(n)`, `VARCHAR(n)`)
  - Bit Strings (`BIT VARYING(n)`)
  - Boolean (`BOOLEAN`)
  - Integer (`INT`, `INTEGER`, `SHORTINT`)
  - Floating-point (`FLOAT`, `REAL`, `DOUBLE PRECISION`, `DECIMAL`, `NUMERIC`)
  - Date and Time (`DATE`, `TIME`)

NOTE: SQL keywords (create and table for example) are not case sensitive.
Named objects (tables, columns etc.) *may* be.

SQL

✓ **Create, Modify, Delete Table**

- Insert Data Into Table
- Single-table Queries
  - The SFW query
  - Useful operators: DISTINCT, ORDER BY, LIKE
  - Handle missing values: NULLs
- Multiple-table Queries
  - Foreign key constraints
  - Joins: basics
  - Joins: SQL semantics
  - Set Operations
  - Inner/Outer Joins
  - Aggregation Queries & Subqueries

SQL: Create

- To create a table, use the **CREATE TABLE** statement
  - Specify the table name, field names and domains

```
CREATE TABLE Customer (
   sin          CHAR(11),
   firstName    CHAR(20),
   lastName     CHAR(20),
   age          INTEGER,
   income       REAL
)
```

```
CREATE TABLE Movies(
    title              CHAR(100),
    year               INT,
    length             INT,
    genre              CHAR(10),
    studioName         CHAR(30),
    studioAddress      CHAR(50)
)
```

SQL: Modify Tables

- Modifying Relation Schemas
  - Delete a relation R: `DROP TABLE R;`
  - Modify Schema of relation R: `ALTER TABLE R`
    - `ADD` followed by an attribute name and its data type
    - `DROP` followed by an attribute name
- Example

```
ALTER TABLE Movies ADD producerCNum INT;
ALTER TABLE Movies DROP studioAddress;
```

SQL: DEFAULT

- Default Values
  - The value used when no other value is known
  - Keyword **DEFAULT**, Value either **NULL** or a Constant
  - Example

```
gender CHAR(1) DEFAULT '?',
birthdate DATE DEFAULT DATE '0000-00-00',

ALTER TABLE MovieStar ADD phone CHAR(16) DEFAULT 'unlisted';
```

SQL: Keys

- Declaring Keys
  - Declare key when attribute listed in the relation schema
    - Can be used only when the key is a single attribute
  - Or add to the list of items declared in schema an additional declaration that an attribute or set of attributes from the key
    - This method should be used if the key consists of more than one attribute
  - Two Declarations
    - Either **`PRIMARY KEY`**
    - Or **`UNIQUE`**
  - Set of attributes ***S*** key for relation ***R***: Two tuples in ***R*** cannot agree on all of the attributes in set ***S***, unless one of them is **`NULL`**. Violating action rejected.

SQL: Keys

- Primary Key Example

```
CREATE TABLE Movies(
    title               CHAR(100),
    year                INT,
    length              INT,
    genre               CHAR(10),
    studioName          CHAR(30),
    producerCNum        INT,
    PRIMARY KEY         (title, year)

);
```

SQL

- Create, Modify, Delete Table

✓ **Insert Data Into Table**

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  - The SFW query
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  - Handle missing values: NULLs

- Multiple-table Queries
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SQL: Insert

- To insert a record into an existing table use the **INSERT** statement
  - The list of column names is optional
    - If omitted the values must be in the same order as the columns

```
INSERT INTO Customer(sin, firstName, lastName, age, income)
        VALUES ('111', 'Sam', 'Spade', 23, 65234)
```

SQL: Modify Records

- Use the **UPDATE** statement to modify a record, or records, in a table
  - Note that the **WHERE** statement is evaluated *before* the **SET** statement
- Like **DELETE** the **WHERE** clause specifies which records are to be updated

```
UPDATE Customer
SET age = 37
WHERE sin = '111'
```

SQL: Delete

- To delete a record use the **DELETE** statement
  - The **WHERE** clause specifies the record(s) to be deleted

```
DELETE
FROM Customer
WHERE sin = '111'
```

- Be careful, the following SQL query deletes *all* the records in a table

```
DELETE
FROM Customer
```


Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

SQL

- Create, Modify, Delete Table
- Insert Data Into Table
- ✓ **Single-table Queries**
  - The SFW query
  - Useful operators: DISTINCT, ORDER BY, LIKE
  - Handle missing values: NULLs
- Multiple-table Queries
  - Foreign key constraints
  - Joins: basics
  - Joins: SQL semantics
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  - Aggregation Queries & Subqueries

Sample Table

Students

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1005	David	SFU	20	3.2
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

The SFW Query

- Select From Where
- To write the query, ask yourself three questions:
  - Which **table** do you want information from?
  - Which **rows** do you want information from?
  - Which **columns** do you want information from?

```
SELECT <columns>
FROM   <table name>
WHERE  <conditions>
```

Columns

- Which **columns** do you want information from?

```
SELECT * FROM Students
```

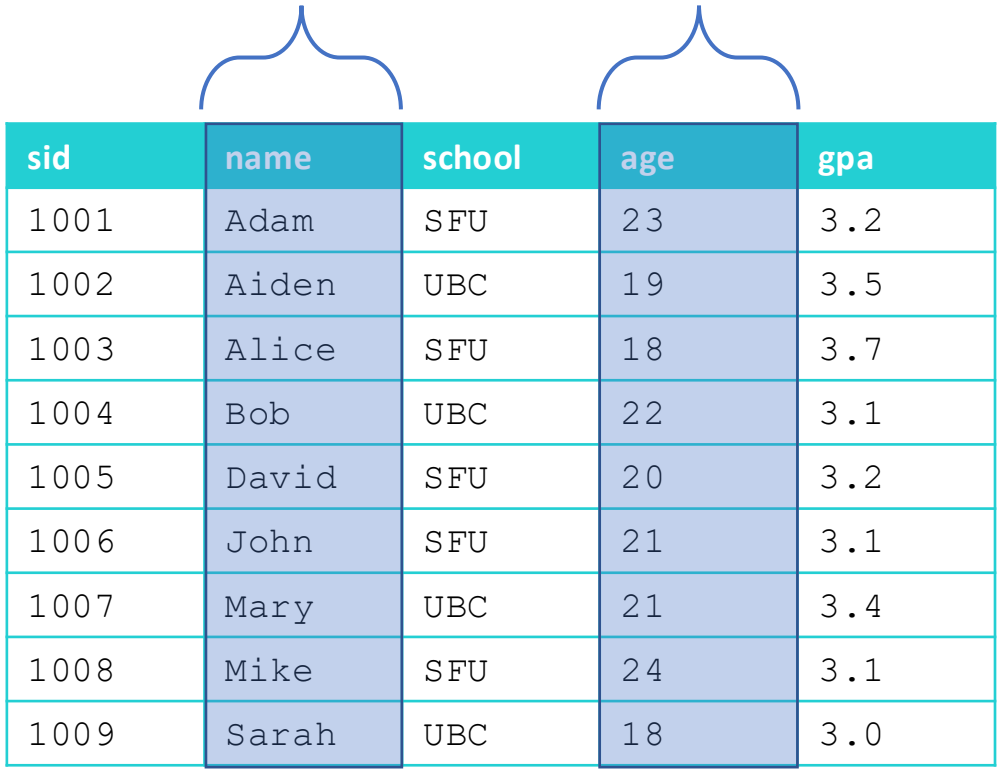
sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1005	David	SFU	20	3.2
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Columns

- Which **columns** do you want information from?

```
SELECT name, age
FROM Students
```



sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1005	David	SFU	20	3.2
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Columns

- Which **columns** do you want information from?

```
SELECT name as studentName
FROM Students
```

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1005	David	SFU	20	3.2
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Columns

- Which **columns** do you want information from?

```
SELECT age * 365 as ageDay
FROM Students
```

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1005	David	SFU	20	3.2
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Conditions

- Which **rows** do you want information from?

```
WHERE gpa >= 3.5
```

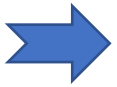
sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1005	David	SFU	20	3.2
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Conditions

- Which **rows** do you want information from?

```
WHERE school='SFU'
      AND gpa >= 3.5
```



sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1005	David	SFU	20	3.2
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Conditions

- Which **rows** do you want information from?

```
WHERE (school='SFU'
       OR school='UBC')
       AND gpa >= 3.5
```

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1005	David	SFU	20	3.2
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Conditions

- Which **rows** do you want information from?

```
WHERE age * 365 > 7500
```

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1005	David	SFU	20	3.2
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Values

- SQL **commands** are **CASE INSENSITIVE**
  - SELECT = Select = select
  - Student = student
  - gpa = GPA
- **Values** are **CASE SENSITIVE**
  - 'SFU' ≠ 'sfu'
- Quotation
  - SQL strings in **single quotes**
    - e.g. name = 'Mike'
    - Single quotes in a string can be specified using an initial single quote character as an escape
      - author = 'Shaq O''Neal'
  - Strings can be compared **alphabetically** with the comparison operators
    - e.g. 'fodder' < 'foo' is TRUE

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Eliminating Repetitive Results

- **DISTINCT**: eliminating duplicates in the query result

```
SELECT school FROM Students
```


school
SFU
UBC
SFU
UBC
SFU
SFU
UBC
SFU
UBC

Versus

```
SELECT DISTINCT school FROM Students
```

school
SFU
UBC

Sorting the Results

- The output of an SQL query can be ordered using `ORDER BY`
  - By any number of attributes
  - In either ascending or descending order

- Default: **Ascending** order

- The keywords `ASC` and `DESC`, following the column name, set the order

Sorting the Results

```
SELECT    name, gpa, age
FROM      Students
WHERE     school = 'SFU'
ORDER BY gpa DESC, age ASC
```

Simple String Pattern Matching

- SQL provides pattern matching support with the `LIKE` operator and two symbols
  - The `%` symbol stands for zero or more arbitrary characters
  - The `_` symbol stands for exactly one arbitrary character
  - The `%` and `_` characters can be escaped with `\`
    - E.g., name `LIKE` 'Michael`\_`Jordan'

```
SELECT * FROM   Students WHERE  name LIKE 'A_d%';
```

Simple String Pattern Matching

- Which names will be returned?

```
SELECT *
FROM   Students
WHERE  name LIKE 'Sm_t%'
```

1. Smit
2. SMIT
3. Smart
4. Smith
5. Smythe
6. Smut
7. Smeath
8. Smt

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

NULL

- Whenever we don’t have a value, we can put a NULL
- Can mean many things:
  - Value does not exist
  - Value exists but is unknown
  - Value not applicable
  - Etc.

- NULL constraints

```
CREATE TABLE Students (
    name CHAR(20) NOT NULL,
    age CHAR(20) NOT NULL,
    gpa FLOAT
)
```


Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

NULL: What will happen?

```
SELECT gpa*100 FROM students

SELECT name FROM students WHERE gpa > 3.5

SELECT name FROM students WHERE age > 19 OR gpa > 3.5
```

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1005	David	SFU	20	NULL
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

NULL: What will happen?

- Arithmetic operations (+, -, *, /) on nulls return **NULL**

```
NULL * 100 = NULL
NULL * 0 = NULL
```

- Comparisons with nulls evaluate to **UNKNOWN**

```
NULL > 3.5 → UNKNOWN
NULL = NULL → UNKNOWN
```

NULL: What will happen?

```
SELECT gpa*100 FROM students

SELECT gpa*0 FROM students

SELECT name FROM students WHERE gpa > 3.5

SELECT name FROM students WHERE gpa = NULL
```

Combinations of true, false, unknown

- Truth values for unknown results

```
TRUE OR UNKNOWN = TRUE

FALSE OR UNKNOWN = UNKNOWN

UNKNOWN OR UNKNOWN = UNKNOWN

TRUE AND UNKNOWN = UNKNOWN

FALSE AND UNKNOWN = FALSE

UNKNOWN AND UNKNOWN = UNKNOWN
```

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Combinations of true, false, unknown

```
SELECT * FROM students WHERE age > 15 OR gpa > 3.5
```

```
SELECT * FROM students WHERE age > 15 AND gpa > 3.5
```

- The result of a **WHERE** clause is treated as **FALSE** if it evaluates to UNKNOWN
  - **WHERE UNKNOWN → FALSE**

Checking NULL Values

```
SELECT * FROM   Students WHERE  age < 25 OR age >= 25
```

```
SELECT * FROM   Students
WHERE  age < 25 OR age >= 25 OR age IS NULL
```

- There are special operators to test for null values
  - `IS NULL` tests for the presence of nulls and
  - `IS NOT NULL` tests for the absence of nulls

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✓ **Multiple-table Queries**

- **Foreign key constraints**
- Joins: basics
- Joins: SQL semantics
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Foreign Key constraints

```
stid REFERENCES sid
```

Students

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1005	David	SFU	20	3.2
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Enrolled

stid	cid	grade
1001	200	A
1001	295	A
1001	250	B+
1002	130	A
1002	125	A+
1003	120	A
1003	125	B
1003	150	B
1004	354	A

?

Foreign Key constraints

```
stid REFERENCES sid
```

Students

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Missing

Enrolled

stid	cid	grade
1001	200	A
1001	295	A
1001	250	B+
1002	130	A
1002	125	A+
1003	120	A
1003	125	B
1003	150	B
1005	354	A

Illegal

Declaring Foreign Key

```
CREATE TABLE Students(
   sid          CHAR(20),
   name         CHAR(20),
   school       CHAR(15),
   gpa          DOUBLE,
   PRIMARY KEY (sid)
);
```

```
CREATE TABLE Enrolled(
    stid  CHAR(20),
    cid   CHAR(20),
    grade CHAR(5),
    PRIMARY KEY (stid, cid),
    FOREIGN KEY (stid) REFERENCES
    Students(sid)
)
```


Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Foreign Key: Insert Operations

- What if we **insert** a tuple into Enrolled, but no corresponding student?
  - **INSERT is rejected!**

Students

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Enrolled

stid	cid	grade
1001	200	A
1001	295	A
1001	250	B+
1002	130	A
1002	125	A+
1003	120	A
1003	125	B
1003	150	B
1015	**371**	**A**

Illegal

Foreign Key: Delete Operations

- What if we delete a student, who has enrolled courses?
  - Options
    - Disallow the delete
      - **ON DELETE RESTRICT**
    - Remove all of the courses for that student
      - **ON DELETE CASCADE**
    - Set Foreign Key to NULL
      - **ON DELETE SET NULL**

Foreign Key: Delete Operations: Restrict

```
CREATE TABLE Enrolled(
       stid          CHAR(20),
       cid           CHAR(20),
       grade         CHAR(10),
       PRIMARY KEY (stid, cid),
       FOREIGN KEY (stid)
        REFERENCES Students(sid)
       ON DELETE RESTRICT
)
```

Foreign Key: Delete Operations: Restrict

Students

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Delete Operation on Students Not Allowed!

Enrolled

stid	cid	grade
1001	200	A
1001	295	A
1001	250	B+
1002	130	A
1002	125	A+
1003	120	A
1003	125	B
1003	150	B

Because sid referenced by stid and `ON DELETE RESTRICT`

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Foreign Key: Delete Operations: Cascade

```
CREATE TABLE Enrolled(
       stid          CHAR(20),
       cid           CHAR(20),
       grade         CHAR(10),
       PRIMARY KEY (stid, cid),
       FOREIGN KEY (stid)
        REFERENCES Students(sid)
       ON DELETE CASCADE
)
```

Foreign Key: Delete Operations: Cascade

Students

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Delete Operation on Students causes delete rows with same sids in stid In Enrolled

Enrolled

stid	cid	grade
1001	200	A
1001	295	A
1001	250	B+
1002	130	A
1002	125	A+
1003	120	A
1003	125	B
1003	150	B

Because sid referenced by stid and `ON DELETE CASCADE`

Foreign Key: Delete Operations: Set NULL

```
CREATE TABLE Enrolled(
       stid          CHAR(20),
       cid           CHAR(20),
       grade         CHAR(10),
       PRIMARY KEY (stid, cid),
       FOREIGN KEY (stid)
        REFERENCES Students(sid)
       ON DELETE SET NULL
)
```

Foreign Key: Delete Operations: Set NULL

Students

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
~~1003~~	~~Alice~~	~~SFU~~	~~18~~	~~3.7~~
1004	Bob	UBC	22	3.1
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Delete Operation on Students causes NULL values in stid on rows with same sids in stid In Enrolled. Still Rejected!

Enrolled

stid	cid	grade
1001	200	A
1001	295	A
1001	250	B+
1002	130	A
1002	125	A+
NULL	120	A
NULL	125	B
NULL	150	B

Because sid referenced by stid and ON DELETE SET NULL

Although satisfies the foreign-key constraint
violates the primary-key constraint
thus deletion operation is rejected

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✓ **Multiple-table Queries**

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Joins

- Single Table
  - **WHERE Conditions**
    - Which **rows** do you want information from?

- Multiple Tables
  - **WHERE Conditions**
    - Which **rows** do you want information from?
    - How to join multiple tables

Join

- Find the names of students with an A+ in 125

```
SELECT  name
FROM    Students, Enrolled
WHERE   sid = stid AND
        cid = 125 AND grade = 'A+'
```

Joining Tables

Rows

Join: Other ways to write the same

```
SELECT name
FROM   Students
JOIN   Enrolled ON sid = stid
WHERE  cid = 125 AND grade = 'A+'
```

```
SELECT name
FROM   Students
JOIN   Enrolled ON sid = stid
       AND cid = 125 AND grade = 'A+'
```

Tuple Variable

```
SELECT name
FROM   Students, Enrolled
```

- What if both tables have an attribute name!?

```
SELECT S.name
FROM   Students S, Enrolled
```

```
SELECT Students.name
FROM   Students, Enrolled
```

SQL

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  - Useful operators: DISTINCT, ORDER BY, LIKE
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✓ **Multiple-table Queries**

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Joins: SQL Semantics

For each tuple `r` in `R` do

    For each tuple `s` in `S` do

        If `r` and `s` satisfy the join condition

            Then output the tuple `<r,s>`

This is called **nested loop semantics** since we are interpreting what a join means using a nested loop

Joins: SQL Semantics

```
SELECT r1.s1, r1.s2, …, rn.sk
FROM   R1 AS r1, R2 AS r2, …, Rn AS rn
WHERE  Conditions(r1,…, rn)
```

```
Answer = {}
for r1 in R1 do
    for r2 in R2 do
        …
            for rn in Rn do
                if Conditions(r1,…, rn)
                    then Answer = Answer ∪ {(r1.s1, r1.s2, …, rn.sk)}
return Answer
```

SQL

- Create, Modify, Delete Table
- Insert Data Into Table
- Single-table Queries
  - The SFW query
  - Useful operators: DISTINCT, ORDER BY, LIKE
  - Handle missing values: NULLs

✓ **Multiple-table Queries**

- Foreign key constraints
- Joins: basics
- Joins: SQL semantics
- **Set Operations**
- Inner/Outer Joins
- Aggregation Queries & Subqueries

Set Operations

- SQL supports union, intersection and set difference operations
  - Called **`UNION`**, **`INTERSECT`**, and **`EXCEPT`**
  - These operations must be performed on **union compatible tables**
    - Two **tables** are **union compatible** if
      - Have same number of attributes
      - Corresponding attributes have same data types

- Although these operations are supported in the SQL standard, **implementations may vary**
  - **`EXCEPT`**
    - May not be implemented
    - May be implemented as **`MINUS`**

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Set Operations

```
SELECT  name
FROM    Students S, Enrolled E1, Enrolled E2
WHERE   S.sid = E1.stid AND S.sid = E2.stid
        AND (E1.cid = 120 AND E2.cid = 125)
```

Set Operations: Union

- Find student names who have taken **120** or **125**.

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

stid	cid	grade
1001	200	A
1001	295	A
1001	250	B+
1002	130	A
1002	**125**	A+
1003	**120**	A
1003	**125**	B
1003	150	B

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Set Operations: Union

```
SELECT name
FROM   Students, Enrolled
WHERE  sid = stid AND cid = 120

UNION

SELECT name
FROM   Students, Enrolled
WHERE  sid = stid AND cid = 125
```

Set Operations: Intersect

- Find student names who have taken both of 120 and 125.

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

stid	cid	grade
1001	200	A
1001	295	A
1001	250	B+
1002	130	A
1002	**125**	A+
1003	**120**	A
1003	**125**	B
1003	150	B

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Set Operations: Intersect

```
SELECT name
FROM   Students, Enrolled
WHERE  sid = stid AND cid = 120

INTERSECT

SELECT name
FROM   Students, Enrolled
WHERE  sid = stid AND cid = 125
```

Set Operations: Except

- Find student names who have taken 125 but not 120.

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

stid	cid	grade
1001	200	A
1001	295	A
1001	250	B+
1002	130	A
1002	**125**	A+
1003	**120**	A
1003	**125**	B
1003	150	B

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Set Operations: Except

```
SELECT name
FROM   Students, Enrolled
WHERE  sid = stid AND cid = 125

EXCEPT

SELECT name
FROM   Students, Enrolled
WHERE  sid = stid AND cid = 120
```


Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Set Operations: Duplicates

- Set operation queries **eliminate duplicates** by default
  - `UNION`, `INTERSECT`, and `EXCEPT`
  - Unlike other SQL operations

- SQL allows duplicates to be retained in these three operations using the `ALL` keyword

Set Operations: Keep Duplicates

```
SELECT name
FROM   Students, Enrolled
WHERE  sid = stid AND cid = 120

INTERSECT ALL

SELECT name
FROM   Students, Enrolled
WHERE  sid = stid AND cid = 125
```

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- Joins: SQL semantics
- Set Operations
- **Inner/Outer Joins**
- Aggregation Queries & Subqueries

Example Tables

Students

sid	name	school	age	gpa
1001	Adam	SFU	23	3.2
1002	Aiden	UBC	19	3.5
1003	Alice	SFU	18	3.7
1004	Bob	UBC	22	3.1
1006	John	SFU	21	3.1
1007	Mary	UBC	21	3.4
1008	Mike	SFU	24	3.1
1009	Sarah	UBC	18	3.0

Enrolled

stid	cid	grade
1001	200	A
1001	295	A
1001	250	B+
1002	130	A
1002	125	A+
1003	120	A
1003	125	B
1003	150	B

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Joins

```
SELECT name, cid
FROM   Students INNER JOIN Enrolled ON sid = stid;
```

```
SELECT name, cid
FROM   Students FULL OUTER JOIN Enrolled ON sid = stid;
```

```
SELECT name, cid
FROM   Students LEFT OUTER JOIN Enrolled ON sid = stid;
```

```
SELECT name, cid
FROM   Students RIGHT OUTER JOIN Enrolled ON sid = stid;
```


Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

SQL

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Simple Aggregation

```
SELECT AGG(column)
FROM   <table name>
WHERE  <conditions>
```

AGG = COUNT, SUM, AVG, MAX, MIN, etc

- Count applies to more than one attribute
- Other aggregations apply to a single attribute

Examples: Simple Aggregation

- `SELECT COUNT(*) FROM Students;`
- `SELECT SUM(gpa) FROM Students;`
- `SELECT AVG(gpa) FROM Students;`
- `SELECT MIN(gpa) FROM Students;`
- `SELECT MAX(gpa) FROM Students;`

Examples: Simple Aggregation

- `SELECT COUNT(DISTINCT gpa) FROM Students;`
- `SELECT SUM(DISTINCT gpa) FROM Students;`
- `SELECT AVG(gpa) FROM Students WHERE age = 19;`

Group By: The Need

- How to get AVG(gpa) for each age?

```
SELECT AVG(gpa) FROM Students WHERE age = 18;
SELECT AVG(gpa) FROM Students WHERE age = 19;
SELECT AVG(gpa) FROM Students WHERE age = 20;
SELECT AVG(gpa) FROM Students WHERE age = 21;
```

Grouping & Aggregation

```
SELECT agg(column)
FROM   <table name>
WHERE  <conditions>
GROUP BY <columns>
```

- How to get AVG(gpa) for each age?
  ```
  SELECT AVG(gpa) FROM Students GROUP BY age;
  ```

Grouping

- How is the following query processed?

```
SELECT age, AVG(gpa)
FROM Students
WHERE gpa > 2.5
GROUP BY age;
```

- Semantics of the query
  - Compute the **FROM** and **WHERE** clauses
  - Group by the attributes in the **GROUP BY**
  - Compute the **SELECT** clause: grouped attributes and aggregates

Grouping

- Special Cases
  - Selecting Attributes
    - Everything in **`SELECT`** must be either a GROUP-BY attribute, or an aggregate
  - Empty Groups

Having

- Specify which groups you are interested in

```
SELECT agg(column)
FROM   <table name>
WHERE  <conditions>
GROUP BY <columns>
HAVING <columns>
```

- **HAVING** clause contains conditions on aggregates.

Example: Having

```
SELECT AVG(gpa), age
FROM Student
WHERE gpa > 2.5
GROUP BY age
HAVING COUNT(*) >= 2;
```

Having: Order of Evaluation

- Create the cross product of the tables in the **`FROM`** clause
- Remove rows not meeting the **`WHERE`** condition
- Divide records into groups by the **`GROUP BY`** clause
- Remove groups not meeting the **`HAVING`** clause
- Create one row for each group and remove columns not in the **`SELECT`** clause

Reference: CMPT 354 (Jiannan Wang- SFU): sfu-db.github.io/cmpt354/

Subqueries

✓**Subqueries**

- `FROM` clause subqueries
- `WHERE` clause subqueries

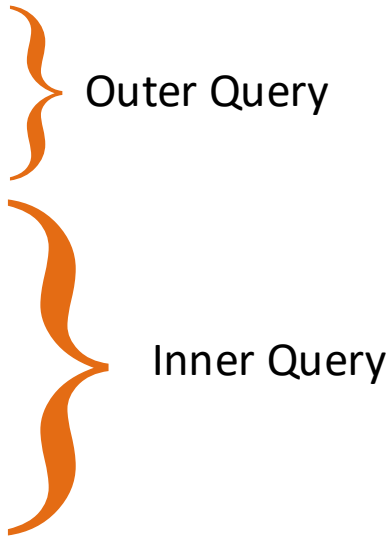
Subqueries

- Need to compute an intermediate table only to use it later in a **SFW**
- Subqueries may appear in
  - A **WHERE** clause
    - Subqueries can return a single constant, and this constant can be compared with another value in a **WHERE** clause.
    - Subqueries can return relations that can be used in various ways in **WHERE** clauses.
  - A **FROM** clause
    - Subqueries can appear in **FROM** clauses, followed by a tuple variable that represents the tuples in the result of the subquery.
- Subqueries may have subqueries, down as many levels as we wish.

Subqueries

- Subquery: a query that is part of another query
- Such inner-outer queries are called nested queries

```
SELECT name
FROM MovieExec
WHERE certNum =
    (  SELECT producerCNum
       FROM Movies
       WHERE title = 'Star Wars'
    );
```



Subqueries in WHERE Clauses

- Subqueries that Produce Scalar Values
- Conditions Involving Relations
- Conditions Involving Tuples
- Correlated Subqueries

Subqueries that Produce Scalar Values

- Scalar: atomic value that can be one component of a tuple

- Comparing the result of a subquery (which is a scalar value) to a constant or attribute in `WHERE` clause

Example: Subqueries that Produce Scalar Values

```
Movies(title, year, length, genre, studioName, producerCNum) StarsIn(movieTitle,
movieYear, starName)
MovieExec(name, address, certNum, netWorth)
```

```
SELECT name
FROM Movies, MovieExec
WHERE title = 'Star Wars' AND producerCNum = certNum;
```

```
SELECT name FROM MovieExec WHERE certNum =
        ( SELECT producerCNum
          FROM Movies
          WHERE title = 'Star Wars'
        );
```

Conditions Involving Relations

- A relation **R** must be expressed as a subquery.

```
IN
NOT IN
EXISTS
NOT EXISTS
ANY
ALL
```

Conditions Involving Relations

- `EXISTS R`: true if and only if **R** is not empty.
- `s IN R`: true if and only if **s** is equal to one of the values in unary relation **R**.
- `s NOT IN R`: true if and only if **s** is equal to no value in unary relation **R**.
- `s > ALL R`: true if and only if **s** is greater than every value in unary relation **R**.
- `s > ANY R`: true if and only if **s** is greater than at least one value in unary relation **R**.

Conditions Involving Tuples

- If a tuple **t** has the same number of components as a relation **R**, then it makes sense to compare **t** and **R** in expressions.

```
SELECT name
FROM MovieExec
WHERE certNum IN
      (     SELECT producerCNum
            FROM Movies
            WHERE (title, year) IN
            (     SELECT movieTitle, movieYear
                  FROM StarsIn
                  WHERE starName = 'Harrison Ford'
            )
      );
```

Correlated Subqueries

- A subquery to be evaluated many times, once for each assignment of a value to some term in the subquery that comes from a tuple variable outside the subquery.

- Example: Finding movie titles that appear more than once

```
SELECT title
FROM Movies Old
WHERE year < ANY
    (       SELECT year
            FROM Movies
            WHERE title = Old.title
    );
```


Reference: *Database Systems: The Complete Book, Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom*

Subqueries in FROM Clauses

- In a FROM list, instead of a stored relation, we may use a parenthesized subquery.

```
SELECT name
FROM MovieExec, ( SELECT producerCNum
                  FROM Movies, StarsIn
                  WHERE title = movieTitle AND
                        year = movieYear AND
                        starName = 'Harrison Ford'
                ) Prod
WHERE certNum = Prod.producerCNum;
```

Acknowledgements

[1] CMPT 354 (Jiannan Wang - SFU)
https://sfu-db.github.io/cmpt354/

[2] Database Systems: The Complete Book, 2nd Edition
Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom
Prentice Hall, 2009

[3] Database System Concepts, Seventh Edition
Avi Silberschatz, Henry F. Korth, S. Sudarshan
McGraw-Hill, March 2019
www.db-book.com

Additional references and resources used in preparation of this course are listed on
https://canvas.sfu.ca/courses/13083/pages/references
or mentioned on each slide.